

Reading the Book of Nature

An Introduction to the Philosophy of Science

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INTRODUCTION

SCIENCE enjoys a lot of respect these days, if not always for the social value of its results then at least for the rigor and precision of its methods. It is an honor to say of a study or an argument that it has been done scientifically. This must be because we think that, in general, science and scientific methods are effective for getting at the truth. If pressed to articulate these feelings we might start by saying that the careful, accountable, and methodical approach of science prevents, as much as possible, the influence of personal bias, superstition, sloppiness, whimsy, sloth, and other human weaknesses that obscure the truth. Thus, when we claim to have done something scientifically, we speak with the authority of a truth-generating enterprise.

There has to be more to it than just being careful (careful to do *what*?) and accountable (accountable by what standards?), and we can appropriately ask why science has its special status as a supplier of knowledge about the world. Thus the unifying question in the pages that follow is, Why should we believe what science says about the world?

This is a question of justification of scientific knowledge. It asks not so much about *what* science claims about the world but more about *how* science proves what it does claim and why it gives us reason to believe that these claims are true. To ask about the justification for knowledge is not an attack on its credibility. There is no implication of skepticism here, and the request for proof is not a veiled suggestion that there is none to give. Instead, the concern for justification represents a requirement of responsibility that is the burden of any purveyor of knowledge. Not just any belief or

statement counts as knowledge, only those that can be justified as likely to be true.

Perhaps, in the case of science, finding the justification is as easy as giving credit for success. Science gains credibility from its phenomenal success. Airplanes fly. Smallpox has been done away with. Humble uranium gets turned into electricity, which in turn keeps the Bud Light cold. This record of success may be enough to give us responsible warrant to believe that science, by and large, gets at the truth about the world, but there are three reasons why this appeal to success is just a suggestive beginning to the analysis of scientific justification.

For one thing, the most interesting science – that is, the sorts of claims whose justification we are most interested in – is about things that nonspecialists can never check for themselves. The currency of science is largely of unobservable entities such as electrons, black holes, DNA molecules, tectonic plates, and the like. We can *see* that airplanes fly, but how do we know that the electrons in an accelerator or in an atom do what the physicists say they do? What counts as success of a theory that is about things that can never be experienced? The criteria and measure of success in this domain are the business of experts, the scientists themselves, and in this sense science is a largely self-regulating business whose success in crucial areas is self-proclaimed.

Again, this is not intended as a threat to the integrity or credibility of science. It is, though, a warrant to investigate further just what it is about science and doing things scientifically that justifies belief in the results being true.

A second reason to be skeptical of the direct association of success as justification is the undeniable fact that the scientific description of the world changes over time. It was once respectable and successful science to describe the universe as being full of ether, and for a very long time the scientific account of the heavens placed the planets and stars in crystalline spheres centered at

and revolving around the earth. We no longer take these once-successful theories to be true. Such a dramatic turnaround should be sufficiently humbling to allow us to ask whether quarks and quasars aren't just today's flash in the pan, which may appear as object lessons in the introduction to a book written years from now on justifying scientific knowledge. An arrogance of the present (yes, but *now* we've got it right) can be avoided by focusing on a general method by which to tell when we've got it right.

The third motivation for looking beyond success is this: Even if these worries about the evaluation of success (it's left up to internal review; it's too ephemeral) could be put aside, there is a deeper question of interest here. What fuels the success? To say that something is done scientifically is not to say simply that it is done with success. Success may be the result, but it is not the methodological structure. What then is this structure, and what about it gives us reason to think it produces truths?

Putting the question in this way motivates the plan of presentation in the following chapters. The first step in the analysis will be an accurate description of the scientific enterprise, focused on the activity of justification of theories rather than their discovery. A clear account of the process of science will then be the evidence for deciding the issue of justification. Once the picture is clear as to how science works, the next step is to ask whether this way of working is conducive to producing the truth. The motivating goal is to be able to evaluate the justification of scientific claims, but first we need to know just what it is we are evaluating.

The specimen under observation here is going to be science at its best. We will not be concerned with fraud in science, episodes such as lying about experimental results, fudging the numbers, and other willful violations of the code of good scientific practice. The topic is the code itself and an assessment of its truth-conduciveness when properly followed. This abstract level of analysis is analogous to a political scientist's study of a social

policy to see if it facilitates a just distribution of social welfare. Of course, there will be cheaters, but before dealing with these problems, the analyst wants to know the worth of the policy when it is properly enforced. Can it deliver as it promises, whether it promises justice or truth, when it is working at its best?

All this talk of justification of scientific knowledge and the scientific method makes it sound as if we have license to generalize about all sciences following a single method at all times. But surely the methods differ from scientific discipline to discipline. From astronomy to zoology, the diverse varieties of sciences study very different kinds of objects and very likely require different ways of theorizing and testing. Even within each domain it is unlikely that there is a single road to the truth, appropriate for all circumstances.

Just as surely, though, all these endeavors have *some* aspect of method in common, and it is this shared methodological ground that is the object of study here. What is it that allows all of these disciplines to be described as being done scientifically and as presenting characteristically scientific evidence? It is at this level of generality that we operate here, and I intend to demonstrate that the common methodological factor in the sciences (when done at their best, don't forget) is important as a truth-indicative ingredient. In other words, working at the high level of generality does not force the discussion to rise above all that is interesting and into the realm of only trivial generalizations. Granted that many of the features common to all the sciences are unimportant to evaluation of justification, there are nonetheless some common features that are significant to justification. But there is no need for you to believe this now, because the demonstration of this point is exactly the burden of what is to follow.

To be blunt, then, I do plan to generalize across the sciences and to proceed as if there is, at some level, a unity of method. I plan to generalize this much and then some by suggesting that the methodological mark of good science is what we take to be

the character as well of good common sense. In other words, as in science so too in life on the street. A close look at science is as a view of our less glamorous reasoning processes under a microscope, in that science is a slowed-down, more open and accountable image of what we normally do in coming to know about the world around us. Good science is an image of good sense.

I do intend to pay attention to a distinction between natural and social sciences, and the discussion here will be limited (though not strictly) to the natural side of the divide. The book of nature is of things without minds or the ideas, intentions, and emotions that are characteristic of minds. The focus on the natural world is not meant to suggest, though, that studying different kinds of objects, nonmental versus mental, necessitates wholly different methods of theorizing and justification. In fact, there is good reason to believe that the methodological model of the natural sciences also fits rather neatly onto the social sciences. I hope to motivate this belief by applying the understanding of natural science to examples of archaeology, indicating that this bridge discipline, with clear affinities to both natural and social studies, is methodologically similar to the natural sciences.

Furthermore, the model that will be used throughout for understanding natural science is one that is most often used in describing social sciences. This is the model of reading a text. Just as science builds models of the world, describing gases as crowds of colliding point-particles, for example, this study of science builds a model of the process of science as being methodologically like reading and interpreting a book, the book of nature. The method by which science comes to understand the natural world is very similar to the method used to understand the meaning of a text written in an unfamiliar language. Understanding the mechanism of nature is like understanding the plot. It is important to realize that this analogy between science and reading is intended to apply only at a methodological level. Though the methods of understanding and justification may be similar between the two

activities, this does not imply that their subject matter is similar as well. In particular, the methodological analogy is not meant to suggest that nature has an author.

The translator of an unfamiliar text comes to understand its meaning through an attention to context. Recurring patterns in the symbols on the page prompt speculation about the meanings of passages, and most importantly, these speculations are testable and amenable to revision. Under the assumption that the text makes sense, that is, that the passages are consistent and they cohere, at least in small sections, into a cogent message, testing a speculated translation of a symbol is done by applying it to other occurrences in other contexts to see if it still makes sense. If the passage comes out as nonsense, the hypothesis must be revised. Through this back-and-forth of suggestion and revision, an understanding of the plot develops and is used to facilitate finding the meaning of passages by giving them a large context into which they fit. Thus the understanding of the whole text guides the interpretation of the parts, and the whole message is itself composed of the meaning of the parts.

This is what happens in reading the book of nature. Instead of marks on a page there are the experiments and observations done by scientists, but like the unfamiliar marks on a page, the observations are meaningless without some prior understanding, if only at the stage of revisable speculation, of what's going on in the story. In the context of science, the observations aren't evidence of anything without a theoretical description of their relevance. For example, the data of a particle detector are interesting only if we know already that a click means that an electron has passed through. Theories, such as the one that describes the link between clicking meters and passing electrons, represent an understanding of general aspects of how the world works and will influence the interpretation of individual pieces of observational evidence. Thus the big picture guides the interpretation of the parts. And of course the theoretical understanding of the big

Introduction

picture is built from and tested against the individual, observational parts.

This reasoning may sound circular, both in the case of translating a text and in that of acquiring knowledge of the natural world, but it is not necessarily a harmful circularity. In both situations, translating and knowing about the world, the project is to describe what is going on behind the scenes. The achievement of science or of translation is to make expansive inferences about things other than but significantly related to what is apparent. It's not interesting simply to describe the appearances, to reproduce or analyze the marks on the page. We want to know the story. We want to know what the marks mean. In science we will never get a look at the answer key, in the form of an authoritative translation of the book or even a dictionary, so we must learn to do the best we can with the available information. The developing theories and theoretically influenced evidence are all we have to go on to draw an accurate description of the world. Of course, somehow this cozy relationship must be regulated to insure a factual reading of the book of nature. What then constrains this process in the direction of truth? That is exactly the right question to ask, and it is time to get on with answering it.

A bit of advice to the reader: The form of this book is true to the message of its content. Individual chapters will make sense and attain their full meaning within the context of the whole work. This is just a friendly warning that the contents are not piecemeal, and parts taken in isolation will convey less information than those parts taken as they participate in the whole. But reading this book will be easier than reading the book of nature, at least in that this one has a lexicon, a glossary of key terms at the end. Terms that are included in this glossary, when first mentioned in the text, will be printed in bold type.

THEORIES

“When *I* use a word,” Humpty Dumpty said in a rather scornful tone, “it means just what I choose it to mean – neither more nor less.”

– Lewis Carroll, *Through the Looking-Glass*

HUMPTY Dumpty knew what he was talking about. We are in control of the meanings of the words we use, if not as individuals then at least as a group, but sometimes words acquire an authority and a persuasion that exceed our understanding of them. Through our profligate use, a word may become influential even when its meaning is unclear and poorly understood. When the link between meaning and emotive influence is strained like this, the authority of language has gotten away from us.

- 、 The word “natural” is a clear example of this. It means roughly that something is not human-made, not artificial, but influence has grown through advertising to suggest that anything natural must be good, wholesome, and healthy. The link between what we mean by “natural” and the authority it has over our behavior is poorly understood, at best. Why are things that are not human fabrications better, across the board, than those that are? More to the point of meaning and understanding, what does it mean to describe a product like shampoo, breakfast cereal, or even rat poison as “all-natural”? And why is it better for me, and the rat too, presumably, if I use all-natural rat poison? The word “chemical” is another case where authority is wielded in the

absence of understanding. Sinister in suggestion though unclear in meaning, it becomes a club of conviction, as we opt for chemical-free products.

A third example of a term whose use and influence are much more common and secure than is its understanding is “scientific.” Treatment of this case will last the rest of the book. It is a persuasive word, and though poorly understood it surely indicates something good in what it describes. Responsible use of the concept and responsible citizenship in a society under the influence of science require clarity in both the meaning of the term and the link between the meaning and its authoritative use.

WHAT THEORIES AREN'T

Still heeding the advice of Humpty Dumpty, we begin our pursuit of clarity in the concept of science with a discussion of the use of a key scientific term, “theory.”

Though unsure of just what it takes to be a **theory** or what it means to be theoretical, people inside and out of science often use the concepts in a vaguely pejorative way, as in “But that’s just a theory!” This suggests a distancing from reality and a lack of proof, an absence of good reason to believe a claim that is *just* a theory. The implicit contrast is between facts, which are true of the world, and theories, which are, well, who knows? It may even be that there is good reason to disbelieve a claim that is merely theoretical. Presentations of the debate between supporters of the theory of evolution and advocates of creation science can exemplify this attitude. (Note the use of the appellation “science” by the latter camp, indicating that this is for them and, they assume, for their audience a thumbs-up word, indicative of good, solid work.) Many representatives of creation science open the attack by describing evolution as just a theory. Good science, the stuff we should be teaching in schools, must be factual science.

— The irony is that the major accomplishments of science, the things that come to mind when we reflect on what good, successful, responsible science is, are all what we describe as theories. Think about it. What has science accomplished? It figured out, for one thing, the phenomenon of universal gravitation, that all massive objects, whether on the earth or in the distant heavens, are forcefully attracted to one another. This describes a variety of happenings in the universe, from the falling of unsupported things around us to the motions of planets and stars, to the expansion of the whole universe. In sum, we, as modern and informed commentators on science, must be referring to the general theory of relativity.

Science also has a practical impact in its application to the difficulties in coping with the world. It has explained the unseen mechanisms responsible for transmission of diseases. Who doesn't believe in the germ theory? And it describes the constituents and events in the microworld with sufficient detail and clarity to support the fairly reliable development of solid-state electronics and, of course, nuclear bombs. We must be talking now of the quantum theory. If you want to know about earthquakes, where, when, and why they occur, and you want a scientific answer, you consult the theory of plate tectonics. And so on.

When you stop talking about theories, you stop talking about science. So, if being scientific is a good thing as far as methodical approach to the truth about the world is concerned, and theories are the primary product of the scientific enterprise, it cannot be that theories are, because they are theories, unlikely to be true or in want of whatever it is that science provides as proof. Being theoretical is nothing to be ashamed about.

Is it something to be proud of? If "theory" is not a term of censure, is it a term of praise, indicative of a credible and comprehensive description of some feature of the world? No, at least not if we pay attention to how the concept is used in science. Consider

two common references to the concept of theory, caloric theory and kinetic theory. Both are descriptions of the phenomenon of heat, though they describe the underlying mechanism in very different ways. The caloric theory has it that heat is a substance (called caloric), a subtle fluid that seeps into and out of things making them hotter and colder, respectively. Being hot is like being wet; only the fluid is different, being caloric instead of water. The kinetic theory, in contrast, claims that heat is not a special kind of stuff but a kind of activity in the molecules of whatever stuff is hot. Nothing flows into a potato to make it hot. Instead, the molecules that make up the potato simply vibrate faster. There is no exchange of materials and, in fact, no such thing as caloric. Clearly, these two theories are in competition as the acceptable description of heat. One is more likely to be true than the other, but neither is more theoretical than the other. The likelihood of each being true is not assessed on the basis of how theoretical each is. The scales are not even coincident. To put it bluntly, it makes as much sense to talk about true theories as it does about false theories. Being theoretical is irrelevant to being true.

It is irrelevant as well to the distinction between being well proven and being new and speculative. The special theory of relativity, with its account of motion, mass, energy, and their interactions in space and time, has passed many scientific tests with excellent scores. It's a well-established theory. Contrast it with superstring theory, a relatively new suggestion about the basic structure of space, time, and everything therein. It is a provocative, though untested, description of the world, and it is a theory. The point is not that these two theories compete (they don't) but that they both fit comfortably within the concept of being a theory even though they differ significantly in their status of being tested.

Perhaps, if "theory" is not laudatory in the sense of being likely to be true, it is a credit to the structure of a claim in that the account is a comprehensive, complete description of some aspect of

the world. It may take several interrelated concepts to cinch up all the loose ends in describing an electron, say, or a tectonic plate, and only the whole package of claims, models, even experimental procedures, counts as a scientific theory. We do sometimes regard theories as complex, multifaceted things, but it is hard to say exactly what must be included to be a complete theory and thus to be a proper unit of scientific evaluation in this sense. It is also unnecessary. Since the project is to assess the justification of scientific claims, it is best to keep a mind open to the possibility of parts of these multifaceted things, that is, individual claims, being justified or not. This advocates an interest in concepts of theories as individual, even isolated, claims and allowance for language such as “the theory that cosmic rays originate in nuclear reactions in outer space.”

This means that you can have theories that are general or specific, comprehensive or selective. Indeed, normal use of the language reflects this openness as we speak of the theory of nuclear fusion as well as a more comprehensive theory of the nucleus. You can have a general theory of the emergence of civilization or a theory of the significance of Stonehenge. You can have a theory that the dinosaurs were done in by catastrophic climatic changes brought on by a huge meteorite. You can even have a theory that it's the guy next door who keeps stealing your Sunday newspaper.

To summarize, none of these features – likelihood of being true, generality, or comprehensiveness – constrains the concept of what it is to be a theory. This generous latitude in the use of the concept devalues its authority. To say of something that it's a theory is neither a credit nor a discredit. There will be good and bad theories in terms of their credibility, but they will be theories nonetheless. What is needed is a way to decide whether a particular claim warrants belief, but simply pointing out that “it's a theory” won't help. We need further information as to whether it's a *good* theory.

Theories

All of this has been about how not to limit or overemphasize the importance of the concept of a theory. It has been about what “theory” doesn’t mean, but surely the concept has some limits. What *does* “theory” mean?

WHAT THEORIES ARE

In answering this question it is best to pay attention to the earlier comment that described theories as the primary products of the scientific enterprise. What then is the goal of science? What is it in the business of producing? But most generally, science aims to provide an account of what’s going on behind the phenomena we experience. By describing objects and events that are not apparent, it tries to make sense of the happenings in the world that are apparent. Science aims to understand the composition and underlying causes of the manifest phenomena and thereby make sense of the phenomenal world we inhabit. This pursuit of understanding could be motivated simply by an urge to satisfy curiosity, or to explain the world, or by a more pragmatic desire to deal with the natural and social environment. Any effort to control the events in the world requires an ability to predict and anticipate happenings and to prevent the undesirable or bring about the desirable. These skills demand an understanding of causes, the unseen mechanisms that determine the events we experience. Thus, whether it is pure or applied, if it’s science it must seek an understanding of the underlying constitution and causes of phenomena.

Science is not after simply a careful, methodical account of the observable world. It is not meant to be but a refined and accurate description of phenomena, the best possible picture of the apparent objects with a precise schedule of their comings and goings. Certainly this sort of detailed and accurate portrayal of the manifest world is important as a necessary means for discovery and testing in science. Discovery and testing of what? Theories. The

accurate observations serve in support of the account science gives of what the world is like in the realm that is too small, too distant, or too far in the past to be experienced.

If science is applied, for example, to the phenomena of acid rain and the dying of forests and lakes, the result is more than a detailed list of the effects. We want to know just what's going on here, and that is exactly what we get. The scientific report is both useful and interesting insofar as it describes the unseen constituents of the problem, the acids and their chemical components, and the basic events of their production and coming together in the atmosphere. This account of what is happening beyond what is directly apparent is the product of science, and it is why we turn to science for help and for understanding.

Earthquakes are another example in which the role of science is to describe what cannot be directly experienced. We can all feel an earthquake, and what we expect of geology is to explain why there are earthquakes and explain it in terms of what it is about the earth that we cannot feel or see that causes the damn things.

The most basic science-classroom demonstrations, those intended to teach what science is all about, are the same sort of thing. My favorite involves rubbing a balloon on someone's hair and then touching the balloon to the wall. It sticks. We start doing science when we ask, What's going on here? But this is not asking for a description of the observable phenomenon. This is asking for an explanation of what makes it happen. What is it about the basic composition of the balloon, the hair, and the wall, and what is it about the event of rubbing that causes the balloon to stick? Maybe a thin skin of old, dry rubber is scraped off when the balloon is rubbed on the hair, exposing the fresh, sticky rubber underneath, and that's what makes it adhere to the wall. Or maybe the important action is at an even smaller scale, and the rubbing actually scrapes microparticles of electric charge from the hair, making the balloon itself electrically charged and attractive to charged particles in the wall. It is this sort of description of

the unperceived factors that are responsible for the phenomena, this kind of behind-the-scenes account of the world, which is the accomplishment of science.

These are theories. A theory is an answer to a “what’s going on here?” question. It is an account of the underlying composition and the unseen causes of the world as we experience it. It is a very liberal concept in that any description of the unexperienced world that is part of what accounts for and helps us understand the experienced world is a theory. There are general theories and very specific theories. Some are well proven, others fairly well disproven, and others still are unproven because they are untested. They have in common the feature of speaking about things beyond the evidence. They take risks in telling us more than we can see for ourselves, and it is exactly this feature that makes them both interesting and in need of justification.

This is not to suggest that there is a clear distinction between theories and facts. It would be misleading to study science by looking for a sharp fact–theory dichotomy, and it would lead to a distorted view of theories, facts, and science in general. This will be made more convincing in Chapter 6, in which the process of scientific observation will be discussed. For now it is enough to be warned that in no aspect of science will the certainty that is characteristic of facts be available. In other words, if indubitability is what it takes to pass from being a theory to being a fact, then there are no graduates and there are only theories. The substantive claims in science are those that speak beyond the obvious, and this brings on a degree of uncertainty. The interesting results are the risky results, and there is no space between trouble and trivia.

The risk is to be minimized, of course, and this is exactly the burden of justification. Theories come in degrees of credibility, and individual theories change over time as they become more or less verified, but the process of justification does not make them more or less theoretical.

It is sometimes helpful to group relevant and cooperating claims into a set. This is what is going on when we talk about, for example, the quantum theory or the big bang theory, multifaceted descriptions of particular features of the world. But responsibility requires that each claim or each model stand the test of justification, and it is unnecessary to focus on whole theories (in this sense of complete accounts) to see if they pass the test wholesale. For this reason, regard theories, insofar as they are the primary subjects of justification, as individual claims.

It will be helpful to point out that there are two kinds of theories within this conception. That is, in understanding the structure of science and its methods of justification, it is useful to distinguish between two kinds of claims about unobservable objects and events. (The use of this term “unobservable” will have to be clarified and somewhat regulated. Between now and Chapter 6 feel free to let common sense be a guide as to what is observable and what is not.)

One sort of theory will be of claims that are only about unobservables. These will be descriptions of the occupants and activities of the unobservable world with no mention of their impact on the observable world. For example, many of the claims made in the physicists’ account of elementary particles make no contact with the world of experience. All baryons (protons, for example) are made of quarks. That describes two kinds of unobservable entities and their relationship of composition. Still more is learned of the affairs of particles from the claim that forces of interaction between them, as occur during collisions, for example, are caused by the exchange of still others, so-called virtual particles. This is a provocative and maybe even accurate account of the microworld, but at this stage it informs of a world isolated from that of experience. The same is true of descriptions of the distant past, reconstructions of the appearance of buildings and pottery, and of political and economic behavior. All claims of this sort describe solely what the describers cannot experience.

• The other type of theoretical claim is needed to break this isolation. This requires claims that are about the interactions between unobservables and observables. In this category are accounts of the events in the past that have influenced the features of the traces found in the present, whether as surviving texts, archaeological remains, or evolved social practices. Other examples include descriptions of the causal impact of the micro-world on the macroworld (and vice versa). These sorts of statements bridge the gap between what we experience, the manifest image of the world, and what we cannot experience but are interested in, the scientific image. And clearly, theoretical claims of this variety will be of essential importance in the understanding of scientific evidence and the process of testing.

Examples of such claims are easy to spot in the experimental activities of scientists. A charged particle, like a proton, will ionize some atoms as it flies by and, in a supersaturated vapor as created in a cloud chamber, visible bubbles will form at the ions. In this way the microparticle leaves its mark on the world of experience and we can keep track of aspects of its behavior. Black holes, since they gravitationally attract light, will bend light rays to the same effect as a lens, causing a visibly altered image of stars and galaxies near their position in the sky. What we cannot see has a noticeable influence on what we can. A ceramic pot, used, broken, and discarded in archaic Greece, gets inadvertently mixed with manure in the animal pens, and pieces are spread with the manure in the fields, only to be recovered and examined by a modern archaeologist who has a theory that the sherd came not from a pot used here in the field, but from one used back at the farmhouse.

These are the imaging theories. Just as the theoretical account of a microscope attests that what is seen is reliably informative of the specimen, these theories of the second kind assure that what we experience, the streak in a cloud chamber or the potsherd, is an informative image of its causal predecessor, the

particle or the pot. It is a distorted image in the sense that protons don't look like vapor trails, and, of course some information about the proton does not show up in the image, but the distortions are taken into consideration. As long as we know that they are there, and it is the business of these theories to point them out, they can be taken into account and they will not deceive.

Our situation with respect to the unobservable world of scientific images has us, as perceivers, at one end of a causal chain. At the other end is the object of interest. The archaic pot, through a sequence of causal interactions including cultural reuse, weathering, and erosion, produces a sherd in a field, and all the archaeologist sees of it is the last step, the sherd. A proton causes ionization, causes bubbles, causes a vapor track, and all the physicist sees is the track. If the sherd or the track are to be evidence of anything, there must be a theory to describe the other end of the causal chain as well as all the intermediate links. Thus both kinds of theories, about unobservables only (such as how archaic Greeks manured their archaic fields, or how protons ionize atoms) and about the relation between unobservables and observables (pots used for cooking are of coarse fabric but with thin walls, and ions initiate the growth of visible bubbles), are used to reconstruct the interactive chain from object of scientific interest to manifest image.

What these claims have in common, what makes them all theoretical, is their reference to what is unobservable. They describe more than what is apparent, and in doing so they draw the question of justification. When is a theory to be believed? Or better, to what degree is it to be believed?

HYPOTHESES AND LAWS

The factors discussed earlier, the generality, comprehensiveness, and degree of verification of a claim, were dismissed at the time

not because they are unimportant but because they were not relevant to the determination of what it is to be a theory. They are nonetheless crucial to an understanding of the process of science, and they coincide with other key concepts used in describing the scientific method.

As pointed out before, some scientific theories have been exhaustively tested and others have not. The former have gone through whatever it is that is the answer to our question of justification. Those that have not or that are just in preliminary phases of testing are hypotheses. To say of something that it is a **hypothesis** is to report on its justification status by describing it as untested or only minimally tested. It is a disclaimer, which says the claim is just suggestive and is asserted without strong commitment. There is nothing irresponsible in suggesting accounts of the world that, as yet, lack justification, provided this status is clearly marked. This is the usefulness of the term “hypothesis.”

Clearly this feature of scientific claims comes in degrees. We will soon find that justification accumulates without ever reaching the stage of certainty. For this reason it is preferable to apply the adjective “hypothetical” rather than the noun “hypothesis” to the description of science and to understand not a dichotomy between what is and is not a hypothesis but the measure of a claim being more or less hypothetical. Science, as a publicly accountable enterprise, requires that its products bear the label of “hypothetical” as a warning if they are tentative and untested. This is a label that wears off slowly, rather than being peeled off at once. The hypothetical status of a theory changes, indicating that this status is not an intrinsic feature, descriptive of what the theory is about or what it says or the form in which it says it. Being hypothetical is a reflection of the theory’s historical relation to the activities (or inactivities) of the scientific community. You can’t tell merely by looking at a theoretical claim just how hypothetical it is. You must be told explicitly.

Sometimes the term “hypothesis” is used in a case-specific way to identify, in a particular experiment, the principal claims being tested. This is to distinguish the subject of testing from the other theories used to account, for example, for the principles of the experimental apparatus and proper conditions. If the cloud chamber is used to investigate the behavior of protons in a magnetic field, the hypothesis is something like this: Protons follow a right-handed spiral trajectory when moving through a magnetic field. There are other theories, as discussed a few pages back, at work in this experiment as well, but they are not the ones the experimenter intends to test. Even with this use of the concept of hypothesis, to be hypothetical is to be in need of testing.

There is another important variation among theories, and that is their degree of generality. Some theories are more general and hence more interesting than others. Theories can be general insofar as they ignore such factors as spatial location (apply anywhere), time (hold anytime), or other features that are irrelevant to the process described. They do this by specifying which factors *are* relevant to the particular behavior. For example, it is true of all unsupported objects very near the surface of the earth that they speed up as they fall. Their speed increases at a rate of 9.8 meters per second for each second they are in free fall. This applies regardless of the color of the object, how it was made, where you bought it, the time of day it is dropped, and so on. These features are irrelevant. What *is* relevant is that the object has mass (though how much mass is not a factor), that it is unsupported, and that it is near the surface of the earth. The theory generalizes over all objects with these properties.

Better yet, or at least more general, is the claim that all massive objects, wherever they are, attract each other with a force that is proportional to the product of their masses and inversely proportional to the square of their distance of separation. This covers a larger class of objects and events, and it specifies fewer relevant properties required for membership in the class. Its contribution

to our understanding of the world is that it identifies the property of things (they must have some mass) relevant to the phenomenon (their attractive force and its resulting motion). It identifies the *kind* of thing relevant to the phenomenon. These theories, which generalize and describe kinds of things, are **laws**. We speak of the law of free fall, itself a chapter in the law of gravity. There are plenty of other examples. Just one more is Coulomb's law, which claims that any electrically charged object exerts a force on any other electrically charged object. The relevant kind of thing here is anything that is electrically charged, and it is linked to the phenomenon of force.

Unlike the feature of being hypothetical, we should be able to tell by looking at the presentation of a theory whether or not it is lawlike. It must be in the form (perhaps implicitly) of a generalization. All *A*'s are *B*. The use of the term "law" here is in one sense much like its use in society. When we say it's the law, we mean it applies to everyone. There are, of course, many important dissimilarities between laws of nature and laws of society. The latter are the convention of (a few or many) people and can be broken or altered at will. Laws of nature, by contrast, are intended to reflect the ways of the world, which are discovered rather than imposed by us and which are largely unbreakable.

We *should* be able to recognize lawlike statements by their generality, but it's not so easy. How can we distinguish real laws from merely accidental associations? Consider the following examples. Pure copper conducts electricity. That's a law of nature. It is an implicit generalization about all samples of copper, and it associates a kind of thing (copper) with a particular behavior (conducting electricity). But here is another generalization that is *not* a law. No samples of pure copper exceed 100 million kilograms in mass. This is general in that it describes all samples of copper and claims they are under 100 million kilos. But this feature of mass is an accidental attribute of copper in that there is nothing, as far as we know, about the nature of copper that puts a limit on the

size of a sample. It turned out, in the chaotic mixing of elements that formed the universe, that no sample of copper that big was formed. But it was just an accident.

And consider the generalization that everyone in this room is over ten years old. That's not a law either. It identifies a particular kind of object (persons in the room), but it is not a *natural* kind. There are no features exclusive to this group that are important to the workings of the world, and there is no causal connection between being in the room and being over ten years old in the way that there is between being electrically charged and being forceful on other charges. Electric charge *causes* the force, but being in the room doesn't cause you to be old. The generalization about the ages of people in the room is most likely just an accidental association. It's not a law. If it were a law it would give me license to make predictions like "The next person to come into the room will be over ten years old." This prediction, however, is unwarranted since being over ten years old was just a circumstantial feature of the people now in the room. The law of conductivity of copper, on the other hand, does warrant the prediction that the next piece of copper will conduct electricity. Laws support predictions; accidental generalizations do not.

So the assessment of whether a certain claim is a law cannot be made simply by checking the form of the claim. It will depend as well on other things we know about the situation, that is, on other theories. It requires an informal preview of what is likely to be relevant to the phenomenon in question. In this way, atomic and solid-state physics indicate that atomic structure, that is, the identity of an element such as copper, is a likely factor in electric conductivity, but nothing links the nature of copper to a limitation on the mass of a sample. Similarly, nothing in my understanding of the room says that being over ten years old is relevant to occupancy. The determination of lawlikeness, as used in the previous examples and generally in our description

of science, is always sensitive to the theoretical context in which the decision is made.

At least the feature of being lawlike is relatively stable over time. This label is not supposed to wear off. Once a law, always a law, almost. In fact, what is assessed to be a law can change as the accepted theoretical description of the world changes. If background theories change, what may once have been assessed as an accidental association could now be understood as a relevant, lawlike connection. In the time of Aristotle, for example, to generalize about motions of things in the sky and on the earth was to group two distinct kinds of phenomena, and any similarities between the two would be viewed as accidental and irrelevant to the understanding of motion. There would have to be two sets of laws, one of celestial motion and another of terrestrial motion. Ever since Newton though, and the discovery that what is relevant to motion is not the terrestrial–celestial distinction but simply the property of mass, it makes perfect sense to generalize over all objects in motion and to regard such generalizations as describing laws of nature.

The lesson, of course, is to resort to an appeal to accident as an explanation of association only as a last resort. But what is important here is simply to appreciate that the concept of law in science is descriptive of the generality and scope of a claim but not its status regarding proof. A law is not an irreproachable, indubitable sort of claim, the deciding word in any dispute. Laws, like any other interesting claim in science, are answerable to the question of justification.

HYPOTHETICAL THEORIES, HYPOTHETICAL LAWS

The three concepts discussed so far, theory, hypothesis, and law, describe different aspects of scientific claims and are not exclusive of each other. They represent different features of claims, just as size, color, and shape represent different features of a

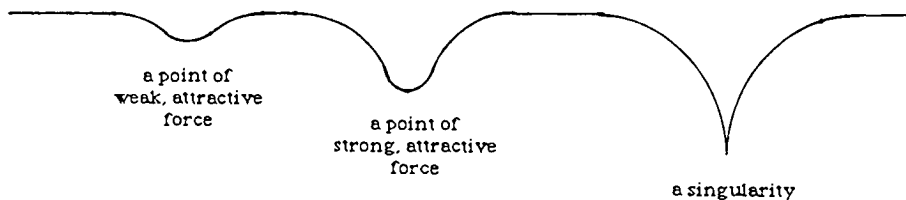


Figure 1.1. The strength of gravitational force is dependent on the curvature of the spacetime manifold. Here, only one of the four spacetime dimensions is shown. A singularity occurs where the curvature is so tight as to buckle the manifold.

physical object, and though there is no sense in describing something as both big and little or both square and round, it makes fine sense to describe it as big and round. Similarly, there is nothing silly or nonsensical in speaking of a hypothetical law or a hypothetical theory. This is no transgression of logic or language.

Here, in fact, is a good example of a hypothetical law, that is, a theory (a claim about events that are not apparent to experience), that is both lawlike (a generalization about a natural kind of object) and hypothetical (not well tested). It is cosmic censorship. The general theory of relativity describes the curvature of spacetime as being associated with gravitational forces. The sharper the curve, the stronger the force. At points called singularities, where the curvature is so sharp that the spacetime actually buckles, the force is so strong as to tear apart elementary particles and to perpetuate events inconsistent with our theories of the normal world (see Figure 1.1). Such a singularity can occur in nature, according to the theory, but it could be surrounded by an event-horizon, a one-way causal shield that prevents both our seeing these crimes against nature and their having any effect on the outside world. This is the situation of a black hole. It is possible as well for there to be singularities without event-horizons, naked singularities, possible at least

according to the general theory of relativity. So cosmologists have considered adding an additional claim to the current description of the universe. Simply, all singularities are in fact surrounded by event-horizons. This abhorrence of nakedness is the principle of cosmic censorship, but it is only at the stage of suggestion, and what is called for at this stage is not instant belief but a further suggestion as to how to test it. It is, in other words, a hypothetical, lawlike theory.

Conversely, there is nothing unreasonable in having a nonhypothetical law or nonhypothetical theory. Snell's law is a nice example. This is the account of the bending of light rays as they pass from one material into another, as from water into air, or air into glass. It specifies not only that the light bends but in which direction it bends and by how much. It all depends on the index of refraction, a property of the material through which the light is traveling. Here then is a theory (insofar as it describes the index of refraction) that is lawlike and very well tested. The best way to put it is that it is very low on the scale of being hypothetical.

The point of this discussion of theories, laws, and hypotheses is twofold. For one, these three concepts are orthogonal; they describe different features of the claims scientists make. In no sense does an idea graduate through the ranks of being hypothetical and then a theory and at last a law. Laws when conceived are hypothetical and are laws nonetheless (nor the more) as testing wears the hypothetical label fainter and fainter. And they are theories all the while.

The second point is that "theory," at least as it is used here, is a very broad term. It is neither laudatory nor pejorative. Since so many things count as being theories they are relatively easy to come by, easy, that is, to discover or create. The hard part is to then justify them, and although there may be lots of theories around, not all of them are good theories. Not all of them warrant our belief.

GOOD AND BAD THEORIES

By good or bad theories I simply mean here that there is or is not good reason to believe that the theory is true. Good theories are justified theories. This, of course, comes in all degrees, all degrees, that is, except perfection. No theory reaches the stage of certainty, of being beyond all doubt and immune from change. But the degrees of justification are exactly what we are interested in here, and we will get on with the project of understanding science when we get on with assessing the virtues theories can have.

So it is not simply the production of theories that makes science distinctive, that endows it with authority and is responsible for its success. Just coming up with theories is not what is scientific about science. Rather, it is the responsibility, declared and dispatched, to separate the good theories from the bad overtly, or at least to scale the better over the worse. It is this process of justification that is at the heart of what it means to be scientific and why that is something to boast about. The understanding of science should focus, then, less on exactly what it is to be a theory and more on the virtues characteristic of a good theory.